COMPUTER VISION TOOL 1.ipynb

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[1]
8s

```
!pip install opencv-python
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
```

[2]
0s

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

[3]
1m

```
from google.colab import files

# Upload image
uploaded = files.upload()
filename = next(iter(uploaded))

# Read image
image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Reduce image size (Improper Sampling)
small = cv2.resize(image, None, fx=0.15, fy=0.15, interpolation=cv2.INTER_AREA)

# Enlarge using nearest neighbor
improper = cv2.resize(small, (image.shape[1], image.shape[0]), interpolation=cv2.INTER_NEAREST)

# Correct using bicubic interpolation
corrected = cv2.resize(small, (image.shape[1], image.shape[0]), interpolation=cv2.INTER_CUBIC)
```

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[5]
✓ 1m

```
uploaded = files.upload()
filename = next(iter(uploaded))

image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

low_light = cv2.convertScaleAbs(
    image,
    alpha=0.4,
    beta=-20
)

gamma = 0.5

table = np.array([
    ((i / 255.0) ** gamma) * 255
    for i in np.arange(256)
]).astype("uint8")

bright_image = cv2.LUT(low_light, table)

lab = cv2.cvtColor(bright_image, cv2.COLOR_RGB2LAB)
l, a, b = cv2.split(lab)

clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)
```

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[5] 1m

```
uploaded = files.upload()
filename = next(iter(uploaded))

image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

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    tileGridSize=(8, 8)
)
```

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[17]
15s

```
from google.colab import files

uploaded = files.upload()
filename = next(iter(uploaded))

image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Add Gaussian Noise
noise = np.random.normal(0, 25, image.shape)

noisy_image = image.astype(np.float32) + noise
noisy_image = np.clip(noisy_image, 0, 255).astype(np.uint8)

# Remove Noise
denoised_image = cv2.GaussianBlur(
    noisy_image,
    (5, 5),
    0
)

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(noisy_image)
plt.title("Noisy Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(denoised_image)
plt.title("Denoised Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Choose Files Screenshot ...142553.png

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[9]
26s

```
plt.axis("off")

plt.tight_layout()
plt.show()
```

Choose Files Screenshot ...115100.png

Screenshot 2026-07-16 115100.png(image/png) - 150775 bytes, last modified: 7/16/2026 - 100% done

```
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```

Low Pixel Resolution | 2-Bit Intensity | 4-Bit Intensity | 8-Bit Intensity

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[19]

```python
from google.colab import files

uploaded = files.upload()
filename = next(iter(uploaded))

image = cv2.imread(filename)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)

faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5,
    minSize=(30,30)
)

for (x, y, w, h) in faces:
    cv2.rectangle(
        image,
        (x, y),
        (x+w, y+h),
        (0,255,0),
        2
    )

    cv2.putText(
        image,
        "Face",
        (x, y-10),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.7,
        (0,255,0),
        2
    )

image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,8))
plt.imshow(image)
```

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[20] 35s

```
from google.colab import files

uploaded = files.upload()
filename = next(iter(uploaded))

# Read image in grayscale
image = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Create different quantization levels
level_2 = (image // 128) * 128
level_4 = (image // 64) * 64
level_8 = (image // 32) * 32
original = image

# Display images
plt.figure(figsize=(16,5))

plt.subplot(1,4,1)
plt.imshow(level_2, cmap="gray")
plt.title("2 Levels")
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(level_4, cmap="gray")
plt.title("4 Levels")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(level_8, cmap="gray")
plt.title("8 Levels")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(original, cmap="gray")
plt.title("Original (256 Levels)")
plt.axis("off")

plt.tight_layout()
plt.show()
```

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[20]
35s

```
from google.colab import files

uploaded = files.upload()
filename = next(iter(uploaded))

# Read image in grayscale
image = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Create different quantization levels
level_2 = (image // 128) * 128
level_4 = (image // 64) * 64
level_8 = (image // 32) * 32
original = image

# Display images
plt.figure(figsize=(16,5))

plt.subplot(1,4,1)
plt.imshow(level_2, cmap="gray")
plt.title("2 Levels")
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(level_4, cmap="gray")
plt.title("4 Levels")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(level_8, cmap="gray")
plt.title("8 Levels")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(original, cmap="gray")
plt.title("Original (256 Levels)")
plt.axis("off")

plt.tight_layout()
plt.show()
```

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[19] 1m

```
plt.figure(figsize=(8,8))
plt.imshow(image)
plt.title("Face Detection")
plt.axis("off")
plt.show()

print("Number of Faces Detected:", len(faces))
```

Choose Files Screenshot ...143525.png

Screenshot 2026-07-23 143525.png(image/png) - 319361 bytes, last modified: 7/23/2026 - 100% done

```
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```
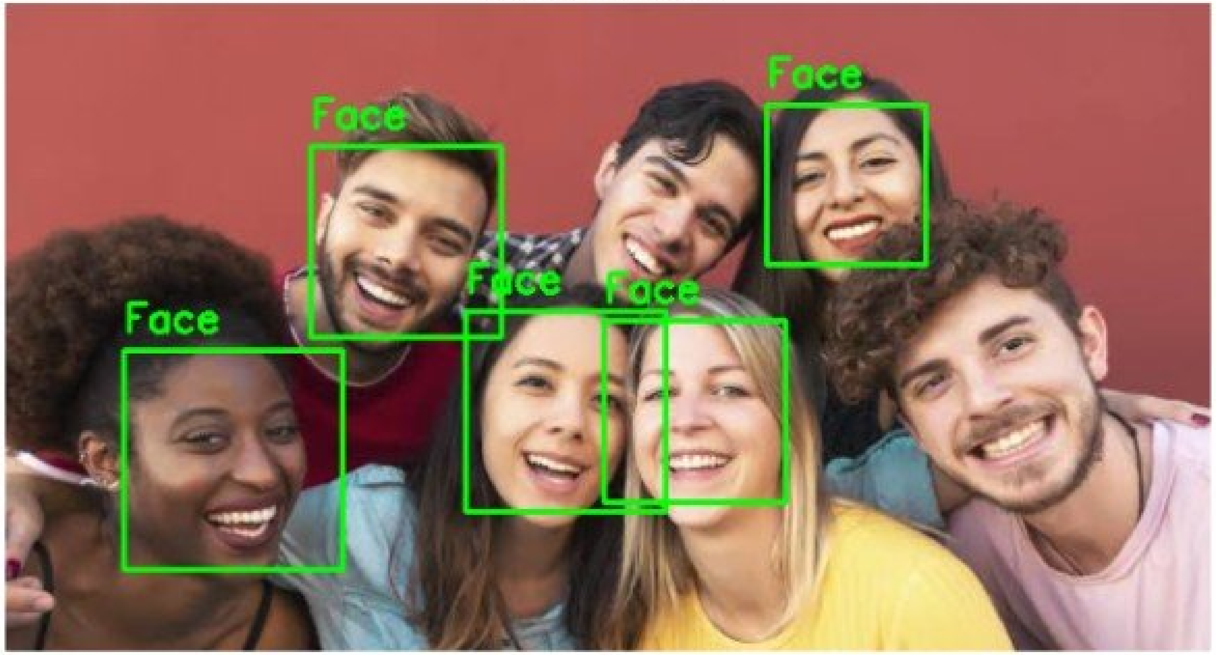

Face Detection




```
Number of Faces Detected: 5
```

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[20] 35s

```
plt.subplot(1,4,3)
plt.imshow(level_8, cmap="gray")
plt.title("8 Levels")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(original, cmap="gray")
plt.title("Original (256 Levels)")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Choose Files Screenshot ...174031.png

Screenshot 2026-07-21 174031.png(image/png) - 156412 bytes, last modified: 7/21/2026 - 100% done

```
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```